Aditya Chauhan

M.Sc. Mathematics · IIT Bhubaneswar
Research: Spectral Graph Theory

+91 94091 64996
aditya1512me@gmail.com
linkedin.com/in/aditya01ad
github.com/aditya01ad
Surat, Gujarat, India

EDUCATION

IIT Bhubaneswar

M.Sc. Mathematics | 2024–2026

CGPA: 8.52 / 10.0

Coursework: Probability & Statistics, Linear Algebra, Real Analysis, Discrete Mathematics, Algebra, Topology, Complex Analysis, Functional Analysis, Numerical Analysis, Optimization Techniques, Measure Theory, Numerical Linear Algebra, PDE, Representation Theory.

VNSGU, Surat

B.Sc. Mathematics | 2021–2024

CGPA: 8.63 / 10.0

Focus: Algebra, Linear Algebra, Real Analysis.

COMPETITIVE EXAMS

GATE 2026 — Mathematics (MA)
✓ Qualified

IIT JAM 2024 — Mathematics
✓ Qualified

JEE Advanced 2020
✓ Qualified

TECHNICAL SKILLS

Active
Python, NumPy, MATLAB, LaTeX, Git

Working Knowledge
Java, C++

AREAS OF INTEREST

- Spectral Graph Theory
- Linear Algebra & Matrix Theory
- Convex Optimization
- Probability & Statistics
- Numerical Methods
- Mathematical Reasoning

LANGUAGES

English — Proficient
Hindi — Native
Gujarati — Native

RESEARCH EXPERIENCE

Graduate Researcher Jun 2024 – May 2026

IIT Bhubaneswar | Supervisor: Prof. Sasmita Barik

Project: *On the Spectral Determination of the Complete Graph with Pendant Vertices*

- Investigated spectral isomorphism in graph theory — whether graph families are uniquely characterised by their adjacency spectrum.
- Derived characteristic polynomials and eigenvalue structures for pendant-attachment constructions via block matrix decomposition.
- Proved the DAS property using spectral invariants (trace, triangle counts, spectral moments) to rule out cospectral candidates.
- Verified theoretical bounds numerically using Python and NumPy.

WHY I FIT AI MATH ROLES

- **Proof writing** — 2 years of rigorous argument construction at IIT level.
- **Error detection** — locating exact failure points in proofs; counterexample construction.
- **Matrix & eigenvalue reasoning** — applied actively in graduate research.
- **Problem formulation** — crafting precise, multi-step, non-trivial problems.
- **Technical communication** — attended two international conferences.

CONFERENCES

- **IWSMGA 2025** — International Workshop on Spectral Methods in Graph Algorithms (attended).
- **ICLAA 2025** — International Conference on Linear Algebra and its Applications (attended).

REFERENCES

Available upon request.